Lesson plan · for the teacher

Teaching it a new tool

AI agents · Level 3 · Lesson 4

Time
40 min

Materials
Two or three class signals pupils already know (hand up, clap twice…); A new signal to teach, such as a hand on the head for “quiet, please”; A tally on the board: old signals and the new one, right or wrong

Learning goals

- Explain fine-tuning: learning from examples.
- Explain forgetting, and why mixing old examples in helps.
- Decide when to fine-tune, and how to test the result.

Activity: Old and new

1. Warm-up (5 min): Show two or three signals the class already knows; pupils respond to each.
2. Teach (8 min): Explain fine-tuning with Darsi Tiny’s numbers: the bus tool from 16 examples; old skills fell from 8 of 8 to 1 of 8 without old examples, and stayed at 8 of 8 with 48 mixed in.
3. Round one (10 min): Teach a new signal (a hand on the head for “quiet, please”) and practise only that one. Then test every signal, old and new, in a random order.
4. Compare (10 min): Tally right and wrong answers for old and new signals on the board. Discuss honestly: people forget far less than a tiny model, but practice still matters.
5. Wrap-up (7 min): Plan round two for another day, with the old signals practised alongside a new one. Then give the exit ticket.

No phones needed: just signals the class knows, one new signal and a tally on the board. If you have a phone, run the teach lab before class to see the real numbers yourself.

Discussion

- When have you learned something new and forgotten something old?
- What would you test before letting pupils use a fine-tuned agent?
- Why might a tool be better than fine-tuning for our school’s room codes?

Exit ticket

- Name one thing fine-tuning can break, and how you would check for it.

Try it at school tomorrow
Teach your class a new hand signal, then test the old ones too. Do it once with practice of the old signals mixed in, and once without, and tally which signals the class still gets right. Then link it to fine-tuning: test the old and the new.

Practice questions · for you

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1 **A colleague fine-tunes the agent on her own examples, and the new answers look great. What should she do before the class uses it?**

- ◯ Test its old skills and its safety too
- ◯ Use it: the new answers look great
- ◯ Fine-tune it again, for longer

2 **Match each way of teaching the bus tool to what the test showed.**

Before teaching	• • • •	Bus 7 of 8; old skills 5 of 8
No old conversations mixed in	• • • •	Bus 5 of 8; old skills 8 of 8
16 old conversations mixed in	• • • •	Bus 0 of 8; old skills 8 of 8
48 old conversations mixed in	• • • •	Bus 7 of 8; old skills 1 of 8

3 **Complete the sentences.**

Fine-tuning changes a model's _____ from examples. With no old examples mixed in, it can _____ its old skills. So always test the old skills and the _____ one.

Word bank: words · new · forget · improve · numbers · tools

4 **After fine-tuning, a model keeps all its old skills, because it only learned something new.**

- ◯ True
- ◯ False

Is there something you would want your school's agent to learn? Would a clearer instruction or a tool be enough, or would it need fine-tuning, and how would you test it?

Check your answers

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1 **Test its old skills and its safety too**

Fine-tuning can quietly break old skills and refusals.

2 **Before teaching — Bus 0 of 8; old skills 8 of 8**
No old conversations mixed in — Bus 7 of 8; old skills 1 of 8
16 old conversations mixed in — Bus 7 of 8; old skills 5 of 8
48 old conversations mixed in — Bus 5 of 8; old skills 8 of 8

New examples alone teach the bus but wipe out old skills.

3 **Fine-tuning changes a model's numbers from examples. With no old examples mixed in, it can forget its old skills. So always test the old skills and the new one.**

Fine-tuning shifts the model's numbers towards the examples.

4 **False**

Learning shifts the same numbers that held the old skills.